ECE 482 Final Project Report

Image Comparator

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1 Introduction

As the application of face detection increases in branches of computer vision such as security, user authentication, social media, and human-computer interaction, the key element to face detection is the image comparator, which assesses visual features across images to determine if they depict the same individual. In our project, we will develop an image comparator that compares two images: our circuit will compare two line-by-line (4 bits at a time for the 4x4 bit images), calculate the number of different bits, and enable another chip to determine whether the two images should be classified as different or the same.

2 Design

2.1 Input Bits

For this image comparator, we initiated the design process with PIPO registers for input X and Y. On the rising edge of CK/4, the 4 bits of X and Y will be stored in $x_3x_2x_1x_0$ and $y_3y_2y_1y_0$, respectively. The PIPO registers consist of four identical registers, wherein the input is stored in the register until the clock signal rises: input D will be given out as output Q only on the rising edge of the divided clock.

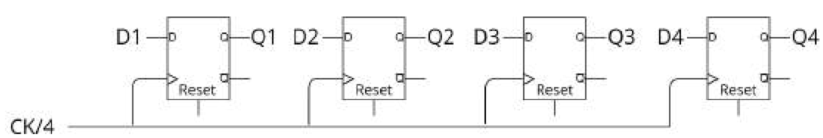

Figure 1: **Schematic of Parallel-In Parallel-Out (PIPO) Register Module**

To divide the master clock frequency by four, we implemented connecting the inverted output of a positive-edge D triggered register as the input of the same positive-edge D triggered register. The output Q results as the master clock frequency divided by two. Then, we connected another positive-edge D triggered register in series to get the quarter of the master clock frequency. To acquire the shift bit signal, we simply added the halved frequency with the quartered frequency together with an OR gate. The OR gate was implemented by complementary logic: we inverted a NOR gate.

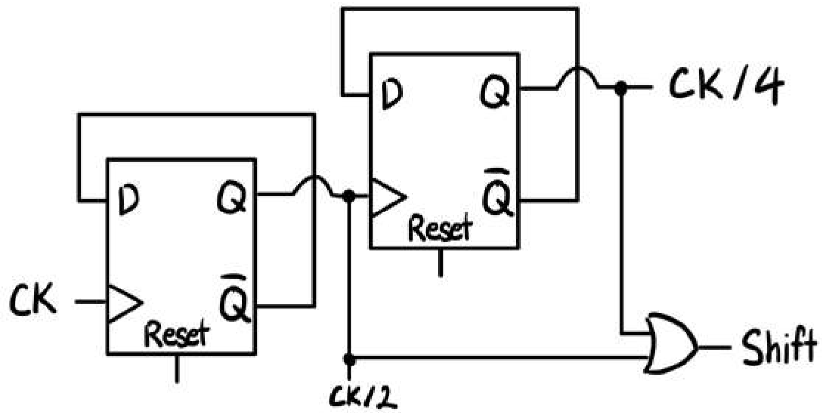

Figure 2: **Schematic of Shift and Clock Divider.** CK/4 frequency gained from two positive-edge triggered D registers in series. Shift signal gained from adding CK/2 and CK/4 frequencies through OR gate.

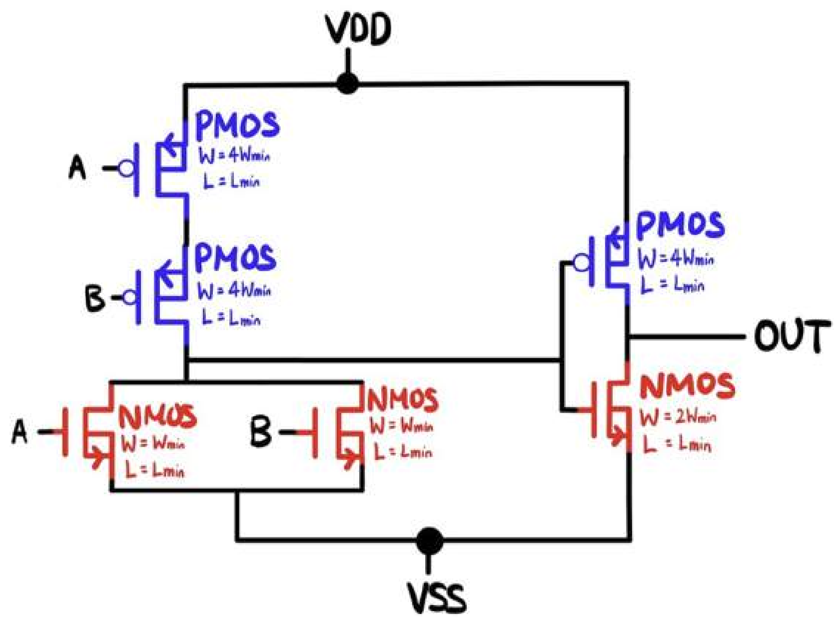

Figure 3: **Transistor Level Schematic of OR Gate.** OR gate was implemented by inverting a NOR gate implemented with complementary logic.

For the registers to be employed in both the PIPO and PISO, we went with C^2MOS based registers in Master-Slave Configuration. A reset functionality was also included with the register. The reset signal (RESET) pulls the output Q to logic low (VSS) via the large NMOS transistor (W = $4W_{\min}$). This larger width ensures faster resetting of the output node, which is critical for applications requiring rapid re-initialization, such as synchronous state machines. The reset functionality is asynchronous, directly affecting the output without requiring a clock. There were several reasons for choosing this design for the registers: The use of C^2MOS transmission gates enables faster operation compared to conventional static CMOS flip-flops. The design minimizes the number of transistors required compared to fully static implementations, which can reduce area and complexity. With all the advantages of this choice, there come certain disadvantages too, like sensitivity to clock skew and race conditions, requirement of careful clock design and charge leakage at dynamic nodes can lead to incorrect operation if the clock period is too long. We made sure to take those into account while designing. Another register design which offers high speeds is True Single Phase Clocking (TSPC) based, however; the master-slave structure in C^2MOS ensures robust edge-triggered behavior and avoids race conditions. TSPC designs can sometimes suffer from incorrect latch behavior under clock jitter or high skew. C^2MOS also avoids issues like charge-sharing and pre-charge failures that can occur in TSPC, especially at very high clock frequencies. The only difference in the registers for the PIPO and PISO modules is the transistor sizing.

With the propagation delay minimized, the two inputs X and Y will be stored in parallel registers, which travel as inputs to the four parallel XOR gates.

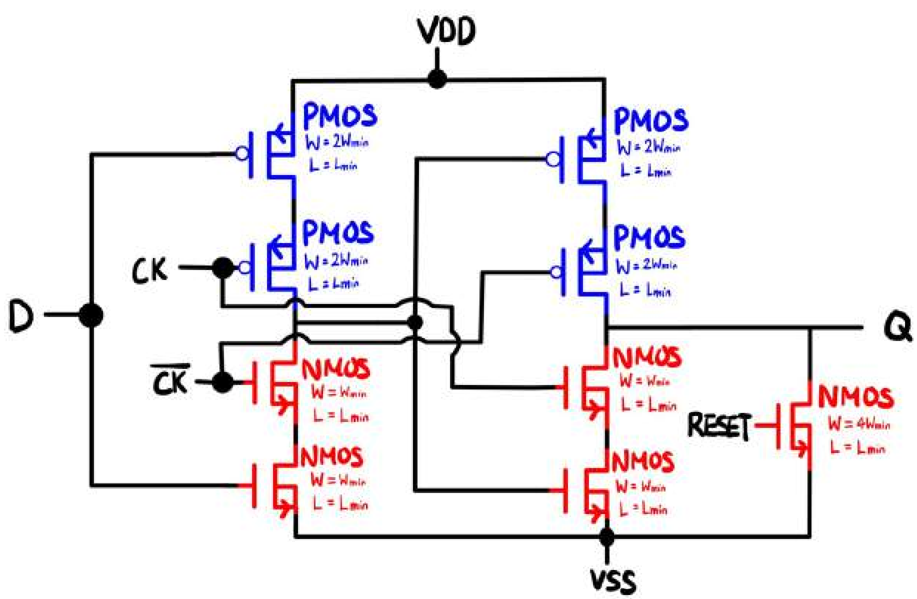

Figure 4: **Transistor Level Schematic of Positive-Edge D Triggered Register With Reset for PIPO Module.**

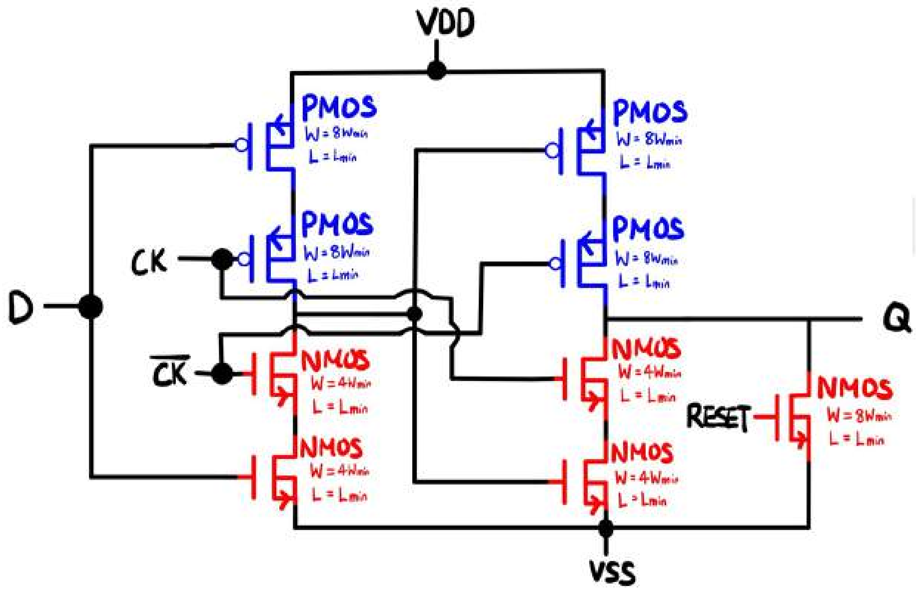

Figure 5: **Transistor Level Schematic of Positive-Edge D Triggered Register With Reset for PISO Module.**

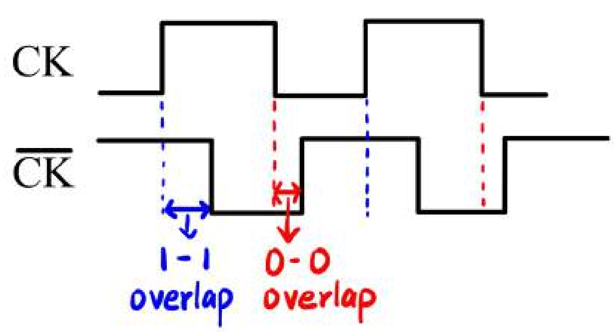

Figure 6: **Diagram of 1-1 Overlap and 0-0 Overlap.** Overlap glitch occurs due to both CK and CK bar being high (or low) from clock skewing.

2.2 XOR

The second part of the design is in charge of comparing corresponding bits through XOR gates, in which the XOR outputs proceed to an adder module. For the XOR gates, we utilized a pass transistor logic with a transmission gate to minimize the number of transistors used. A total of 6 transistors are utilized (4 to create the XOR, 2 to invert one input), which differs from the 8 transistors required for a complementary CMOS logic for the same XOR gate. Using transmission gates for XOR circuits offers distinct advantages over static CMOS implementations. Transmission gates, which utilize both NMOS and PMOS transistors, ensure that signals are passed with minimal degradation, enabling full voltage swing and robust logic levels. This makes them superior to pass-transistor designs that can suffer from voltage threshold drops. Compared to static CMOS, transmission gates require fewer transistors for implementing XOR functionality, resulting in reduced area and lower capacitance, which in turn improves speed. Additionally, transmission gates are highly versatile, allowing bidirectional signal flow, which is beneficial in multiplexing and other advanced logic applications. While static CMOS XOR gates provide better noise immunity in certain scenarios, transmission gates achieve a balance of compactness, speed, and signal integrity, making them an efficient choice for XOR circuits in modern digital systems.

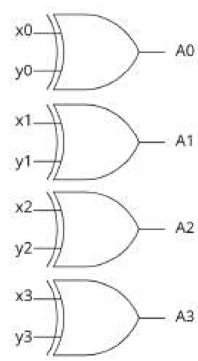

Figure 7: **Schematic of XOR2 Gates for Corresponding Input Bits**

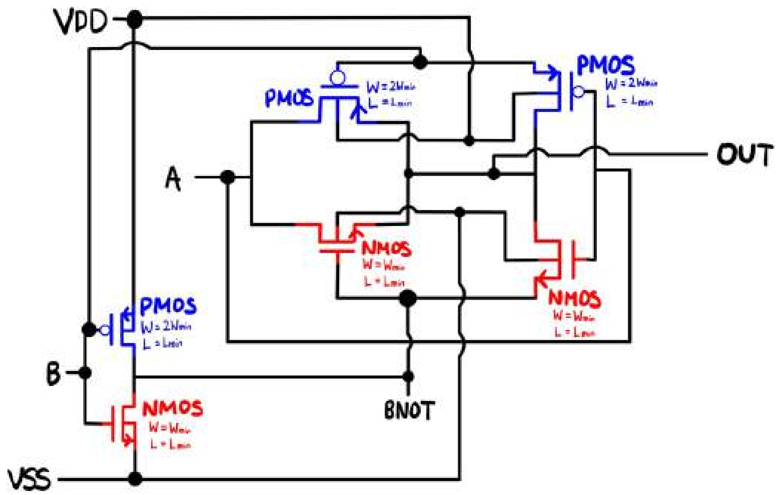

Figure 8: **Transistor Level Schematic of XOR2 Gate**

The XOR gates will have outputs of A_0, A_1, A_2, and A_3, in which A_0 is equal to x_0 XOR y_0, A_1 is equal to x_1 XOR y_1, and so on.

2.3 Half Adder

The outputs of the XOR gates will then serve as inputs for half adders. Half adders will have two outputs: S and C_{OUT}. Output S will be the output of two inputs through an XOR gate, while C_{OUT} will be the output of two inputs through an AND gate.

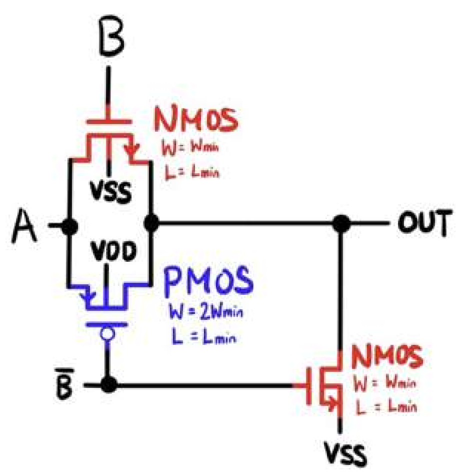

Figure 9: **Transistor Level Schematic of AND2 Gate**

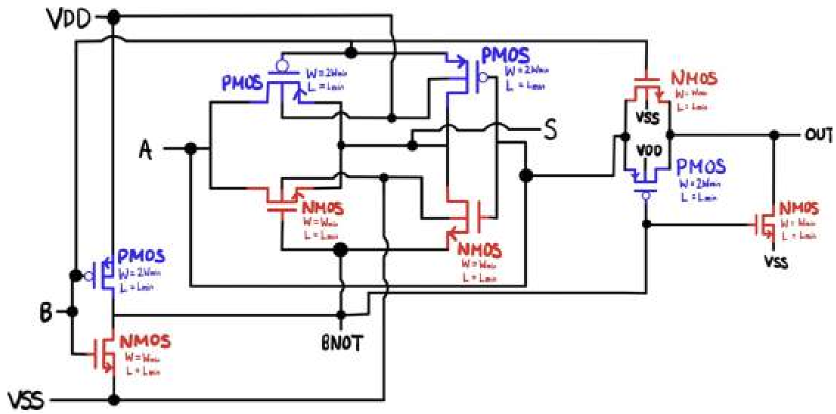

Figure 10: **Transistor Level Schematic of Half Adder**

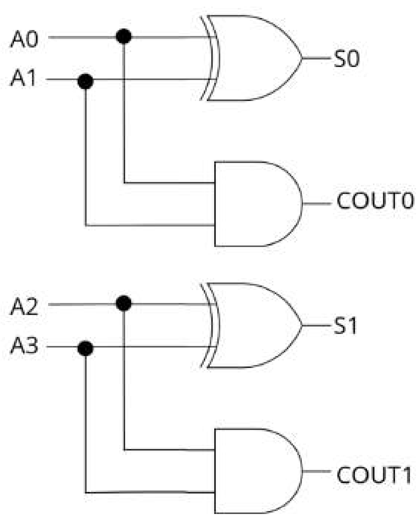

Figure 11: **Schematic of Two 1-bit Half Adders**

A total of three half adders will be utilized to determine the least significant bit r_0: Bits A_0 and A_1 will be inputs for one half adder, and bits A_2 and A_3 will be for the other half adder. The S bits for these half adders will act as the input of a third half adder, in which the output S for this half adder will be r_0.

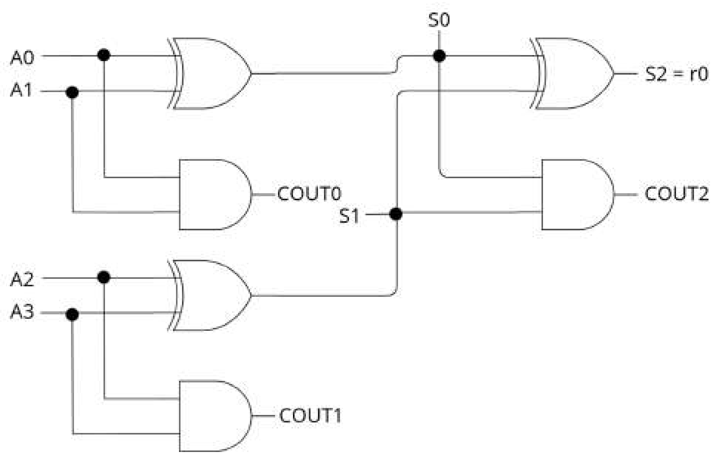

Figure 12: **Schematic of three Half Adders To Determine Bit r_0**

2.4 Full Adder

The remaining three C_{OUT} bits from the half adders will become inputs for a 1-bit full adder. The full adder also produces two bits S and C_{OUT}, however, it takes in three input bits. Output S will only require two XOR gates to XOR all three inputs. This output bit will correspond to bit r_1.

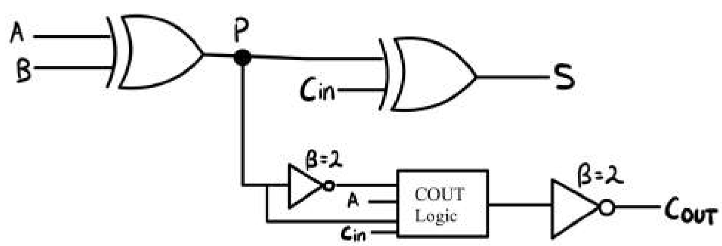

Figure 13: **Schematic of Full Adder with Inputs from Half Adder Outputs.** Inputs A, B, and C_{in} corresponds to C_{OUT} values from half adders. Output S corresponds to bit r_1, and output C_{OUT} corresponds to bit r_2.

However, C_{OUT} of the full adder requires more transistors to implement. We used 8 transistors to produce an inverted C_{OUT}, and another 2 transistors for the actual C_{OUT}, so a total of 10.

This implementation uses a smaller number of transistors (20 transistors) than some of the other static circuits we studied. We concluded that a complementary static CMOS implementation that generates the same two signals C_O and S will have 26 transistors. A mirror adder that generates C_O and S also uses 26 transistors. The circuit shown above reduces the area and the power consumption of the adder. Furthermore, it minimizes the number of devices in the critical path from C_i to C_O.

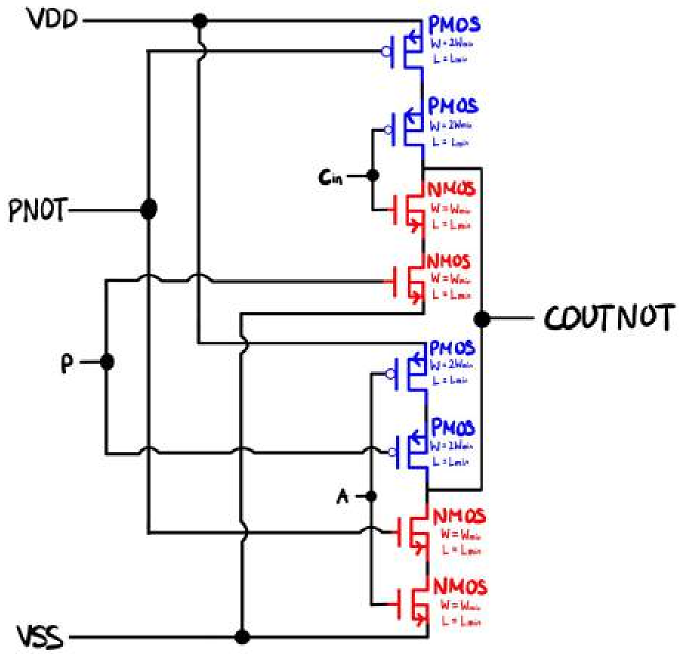

Figure 14: **Transistor Level Schematic of COUT Logic**

2.5 PISO Module

The third part will be the PISO (Parallel-Input Serial-Out) module. This module consists of two components: MUX and the aforementioned positive-edge D registers. These shift registers will shift bits on the basis of the clock frequency (CK) and output data as Serial_OUT. The MUXes were designed in a similar manner to how an XOR gate would be designed: transmission-gate-based adder. While an XOR gate requires one transmission gate, the MUX will require two transmission gates for two inputs.

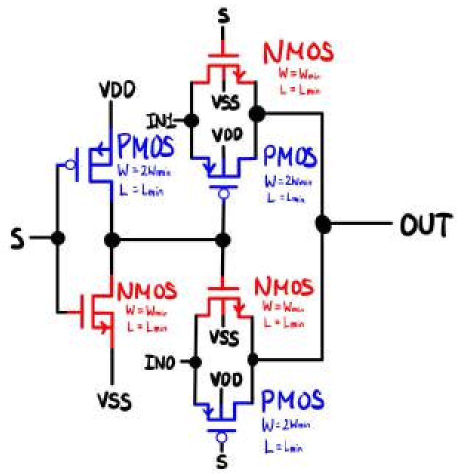

Figure 15: **Transistor Level Schematic of MUX2**

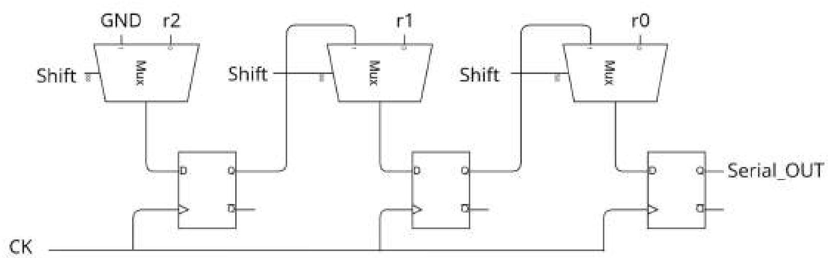

Figure 16: **Schematic of PISO Module**

As the name suggests, bits r_2, r_1, and r_0 are parallel inputs to the three MUXes. When the shift bit is low (= 0), these bits will be stored in the flip flops, waiting for the shift bit to rise (= 1). Once the shift bit equals 1, the data bits stored / loaded in the flip flops will "shift" to the next component, where bit r_2 will become an input for the MUX containing r_1, bit r_1 will become an input for the MUX containing r_0, and r_0 will be the output of the module. These all occur on the rising edge of the clock frequency CK. For example, when the clock is on its rising edge on three consecutive occurrences while the shift bit is high, Serial_OUT will yield r_0, r_1, and r_2 in order.

2.6 Level Shifter

The last part of the design consists of the level shifter and the off chip driver, which exists for the image comparator to decide whether the two image input bits are same or different. The level shifter will utilize VDDIO instead of VDD, for the off chip driver uses a higher voltage, compared to the logic used for the image comparator. To convert VDD (= 1.1 V) to VDDIO (= 1.8 V), the level shifter is designed with a Differential Cascode Voltage Switch Logic (DCVSL), which poses advantages with having a lower propagation delay due to fast charging / discharging speed, when compared to a design implemented only with static CMOS.

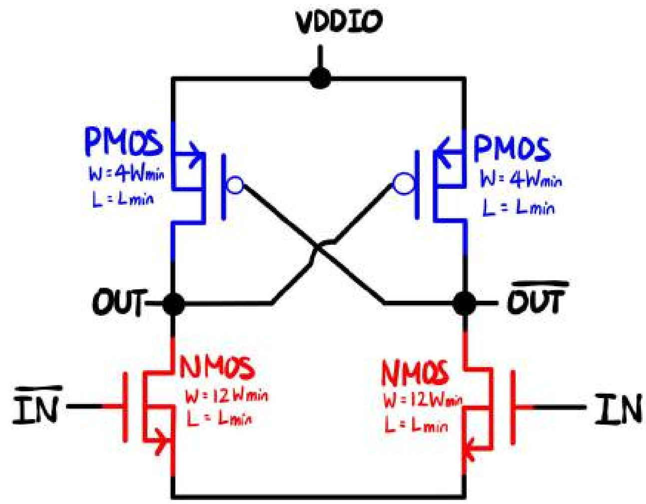

Figure 17: **Transistor Level Schematic of Level Shifter Using DCVSL**

2.7 Off Chip Driver

The off chip driver will be a chain of inverters to drive the load capacitance of 10 pF. From process parameters and simulations, values $t_{p0} = 6.7$ ps, $C_{g.min} = 1.54$ fF, and $\gamma = 0.9$ were gained. Since the output signal of the complement of the level shifter will be used, the total number of inverters added should be odd (N = even). Below demonstrates how the fanout of the inverters were chosen:

$$f_{g0} = exp(1 + \frac{\gamma}{f_{g0}})$$

$$f_{g0} = 3.51$$

$$N_{opt} = \frac{ln(\frac{10pF}{4*1.54fF})}{ln(3.51)} = 5.89$$

With N = 6 and S = 4, we can calculate the fanouts for the inverter chain:

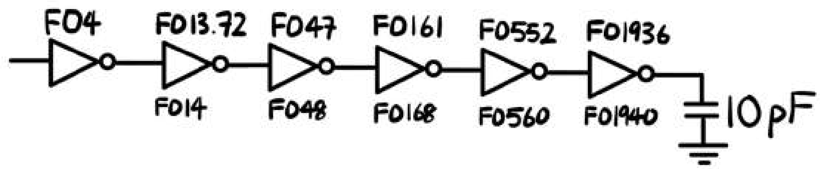

Figure 18: **Schematic of Inverter Chain for Off Chip Driver.** The values above the inverters are theoretical values, while the values below the inverters are values used in design.

From Figure 16, the fanout values we utilized were different from what was theoretically calculated. The difference in this comes from the simplicity in designing our inverters: fanouts in multiples of 20 were easier to design for the large fanout inverters. In addition to simplicity, we wanted to ensure that every two inverters were sized to have a common factor between them such that the height of the cells were matched.

The chip driver sizes were manipulated to minimize propagation delay in case the circuit was capable of running at higher frequencies. Inverters with FO $\sim$ FO3 inverters also meet this requirement in our design, allowing a smaller chip driver with less power consumption. The drawbacks from this approach was that, however, the final design required a larger area which led to higher power consumption from the resulted larger inverters.

	PMOS	NMOS
XOR2	$2W_{min}/L_{min}$	W_{min}/L_{min}
NOR2	$4W_{min}/L_{min}$	W_{min}/L_{min}
Inverter for OR2	$4W_{min}/L_{min}$	$2W_{min}/L_{min}$
MUX2	$2W_{min}/L_{min}$	W_{min}/L_{min}
Positive Edge Triggered D Register (PIPO)	$2W_{min}/L_{min}$	W_{min}/L_{min}
NMOS For Reset of Register (above)		$4W_{min}/L_{min}$
Positive Edge Triggered D Register (PISO)	$8W_{min}/L_{min}$	$4W_{min}/L_{min}$
NMOS For Reset of Register (above)		$8W_{min}/L_{min}$
COUT Logic	$2W_{min}/L_{min}$	W_{min}/L_{min}
Inverter for Clock frequency	$7W_{min}/L_{min}$	$3W_{min}/L_{min}$
Inverter for PISO	$6W_{min}/L_{min}$	$3W_{min}/L_{min}$
Inverter for PIPO	$2W_{min}/L_{min}$	W_{min}/L_{min}
Level Shifter	$4W_{min}/L_{min}$	$12W_{min}/L_{min}$

Table 1: **Transistor Sizing for Different Components Used in Design**

Maximum Frequency	2.33 GHz
Average power drawn from VDD at MIN	193.45 uW
Core Supply Figure-Of-Merit: Power / Frequency	83.18 fJ
Average power drawn from VDDIO at MIN	11.35 mW
IO Supply Figure-Of-Merit: Power / Frequency	4.88 pJ
Area of Top Level Layout (um x um)	24.77 * 54.46 um^2
Latency (number of CK cycles)	3 $\sim$ 4 cycles

Table 2: **Required Values For Project Design**

3 Circuit Functionality Demonstration

3.1 Corner TT

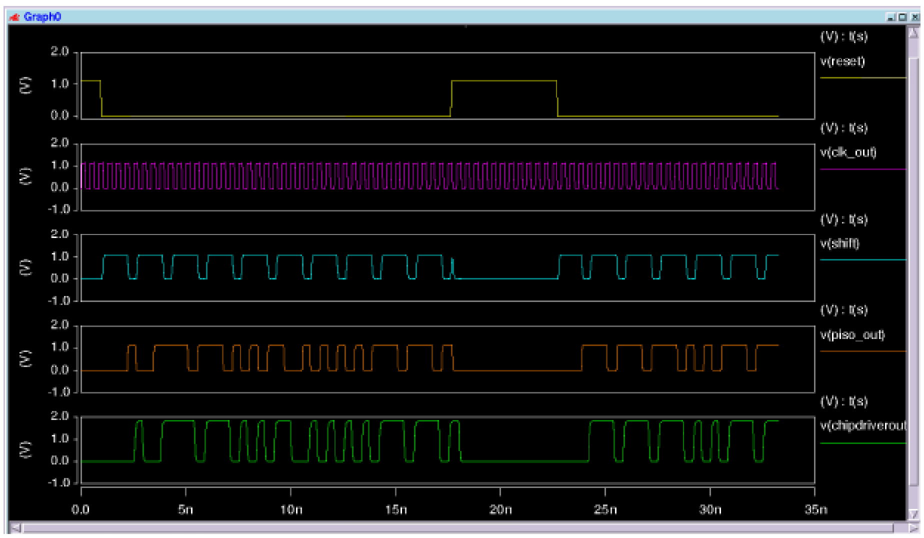

Figure 19: **Waveform of TT Corner at Target Frequency.** Waveform includes CLK_out, Reset, Shift, Serial_OUT (Piso_out on our waveform), and Driver_OUT (Chipdriverout on our waveform) signals.

3.2 Corner SF

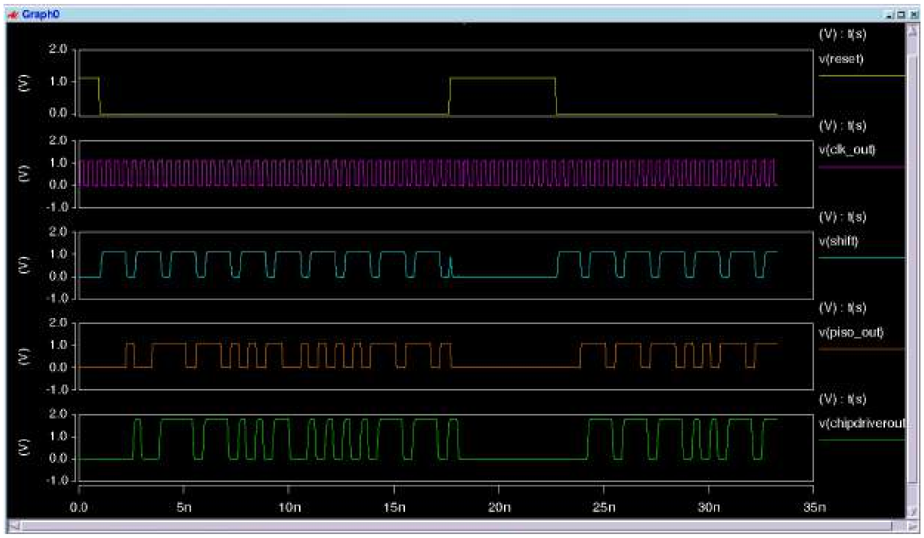

Figure 20: **Waveform of SF Corner at Target Frequency.** Waveform includes CLK_out, Reset, Shift, Serial_OUT (Piso_out on our waveform), and Driver_OUT (Chipdriverout on our waveform) signals.

3.3 Corner FS

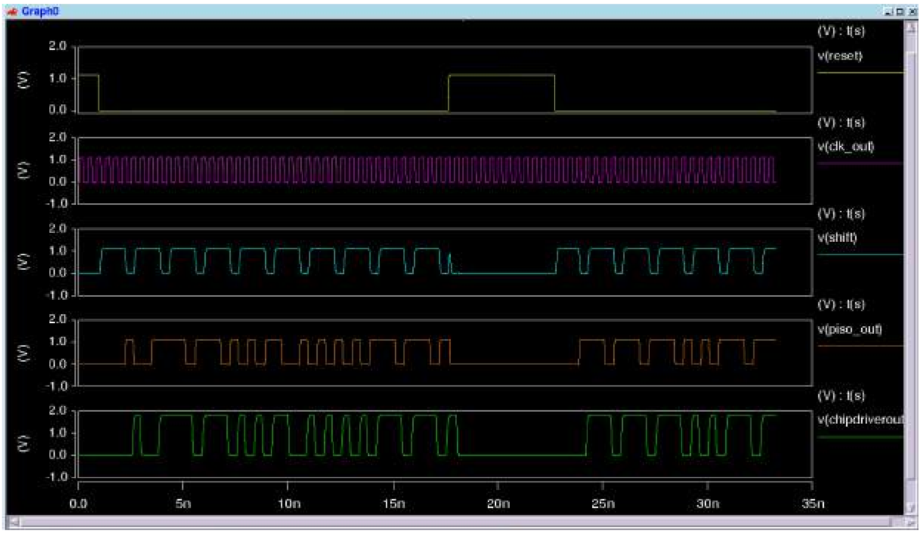

Figure 21: **Waveform of FS Corner at Target Frequency.** Waveform includes CLK_out, Reset, Shift, Serial_OUT (Piso_out on our waveform), and Driver_OUT (Chipdriverout on our waveform) signals.

3.4 Corner FF

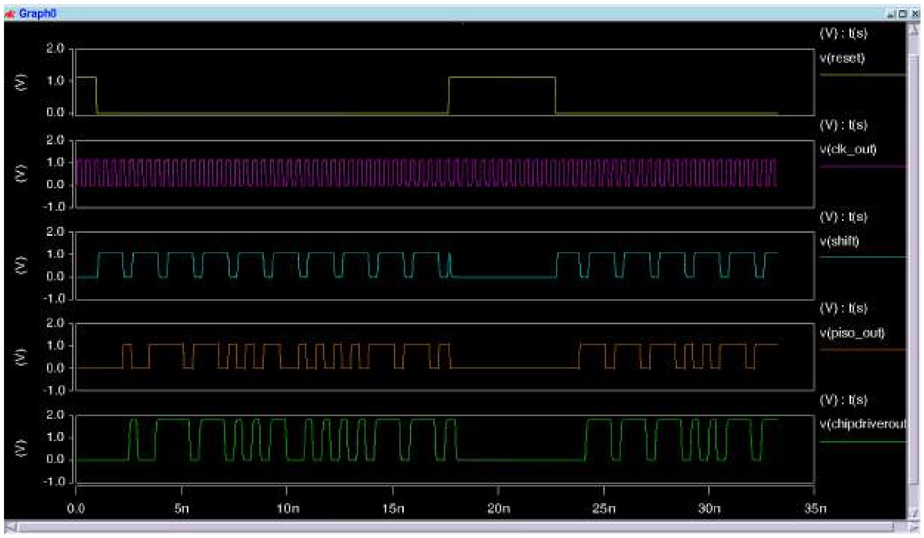

Figure 22: **Waveform of FF Corner at Target Frequency.** Waveform includes CLK_out, Reset, Shift, Serial_OUT (Piso_out on our waveform), and Driver_OUT (Chipdriverout on our waveform) signals.

3.5 Corner SS

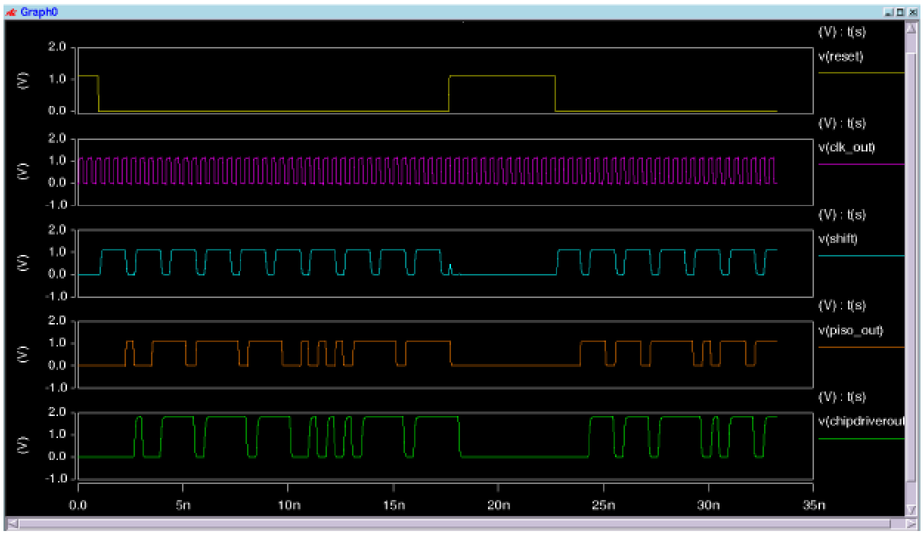

Figure 23: **Waveform of SS Corner at Target Frequency.** Waveform includes CLK_out, Reset, Shift, Serial_OUT (Piso_out on our waveform), and Driver_OUT (Chipdriverout on our waveform) signals.

Our circuit did not function at the target frequency for the SS corner, so we had to reduce the frequency to 2.33 GHz (period = 0.430 ns).

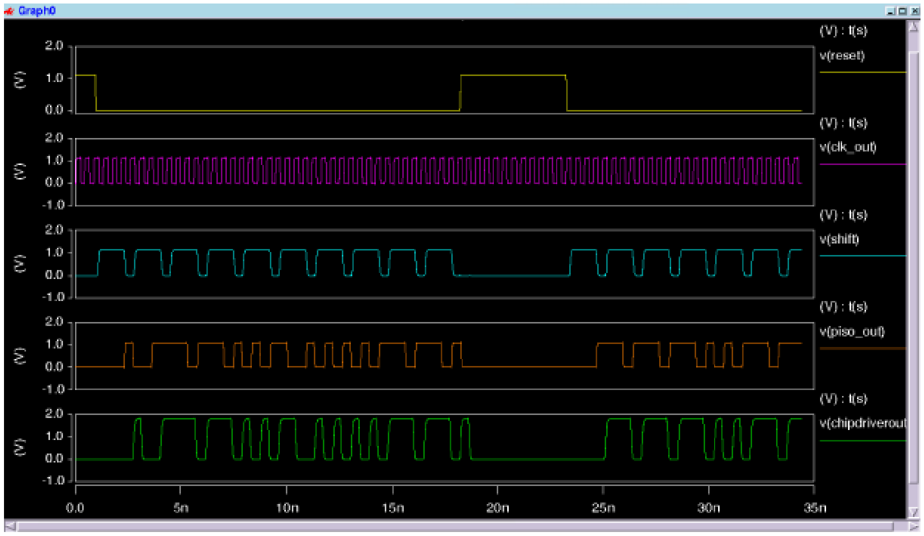

Figure 24: **Waveform of SS Corner at Maximum (Functioning) Frequency.** Waveform includes CLK_out, Reset, Shift, Serial_OUT (Piso_out on our waveform), and Driver_OUT (Chipdriverout on our waveform) signals.

4 Discussion of Results

For the layout, we started by analyzing outputs part by part. The PIPO registers, XOR gates, and the adder modules functioned well at the desired frequency. The PISO module, however, initially did not pass the SS and FS corners, so we had to modify the layout. Initially, the NMOS of the PISO sizing was equal to W_{min}, which was not strong enough to pull down the output bit r_0: When $r_2 r_1 r_0$ should output 010, we would have a result of 011. Thus, through numerous simulations, the NMOS sizing was increased to 4 * W_{min} to be strong enough to pull down the signals, enabling the bits to be pulled down. Since the NMOS sizing changed, the PMOS sizing also changed from 2 * W_{min} to 8 * W_{min} to match the beta factor of 2.

After modification and simulations, the PISO module passed all corners at the desired maximum frequency. When all of the parts were put together, all corners except the SS corner passed at the maximum frequency (2.4 GHz). The SS corner could pass only up to a frequency of 2.33 GHz.

To tackle this problem, we tried all different methods to pass the SS corner at the target frequency of 2.4 GHz; however, we failed to do so. In fact, the same problem that occurred at the PISO module occurred for the entire layout: the pull down network was not strong enough to pull signals down for desired outputs. We concluded that the buffers connected in between modules did not reduce the propagation delay, but increase it instead. Thus, we decided to remove buffers that were in between the clock divider and the PIPO module, and in between the shift signal and the PISO module. This allowed our design to run at a frequency of 2.33 GHz, while it previously could only work up to 2.1 GHz.

Although our design did not work at the maximum target frequency, we may have further reasons why the SS corner did not pass. The SS process corner is denoted as the slow-slow corner, in which the carrier mobilities of the MOSFETs are slower than usual. The decreased speed in the MOSFETs can be mainly affected by high temperatures and high threshold voltages.

With an increase in temperature, the carriers within the devices would undergo more frequent lattice vibrations, thus leading to phonon scattering, a phenomenon in which lattice deformation leads carriers to travel in unwanted directions. Also, when the threshold voltages increase, the devices will require a greater voltage to turn "on" and conduct, limiting the drive current when the devices are actually on.

These two factors may lead to carrier mobility degradation, which may be one of the multiple limiting factors why our layout did not function at the target frequency.

5 Layout Top View

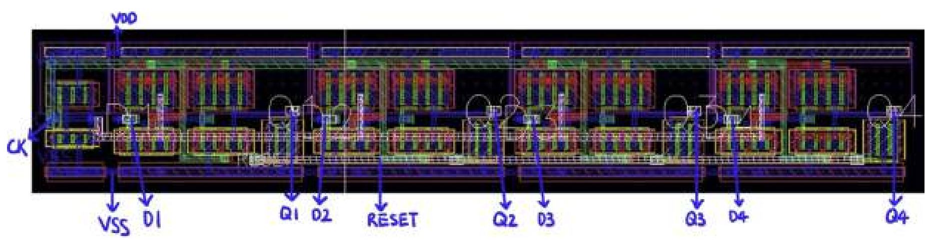

Figure 25: **Layout of PIPO Module**

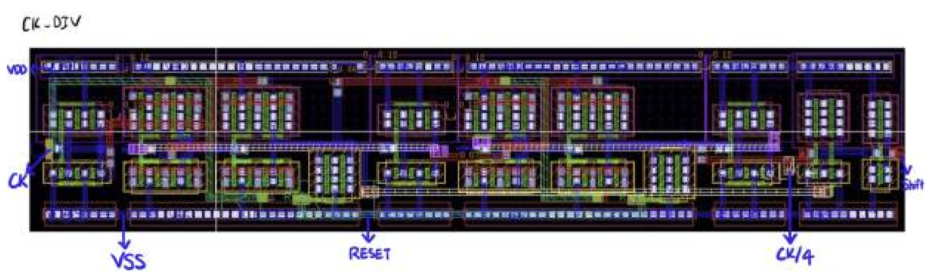

Figure 26: **Layout of Clock Divider and Shift Signal**

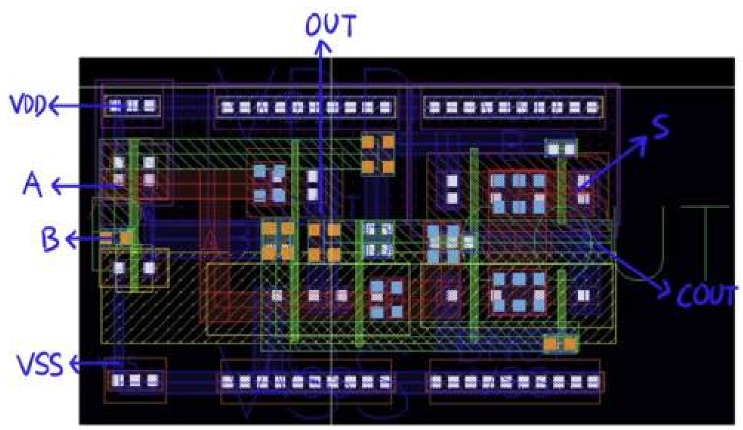

Figure 27: **Layout of Half Adder**

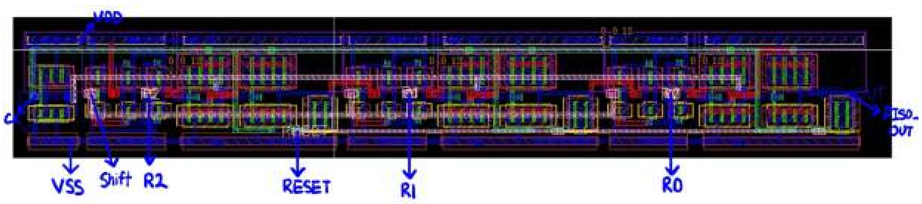

Figure 28: **Layout of PISO Module**

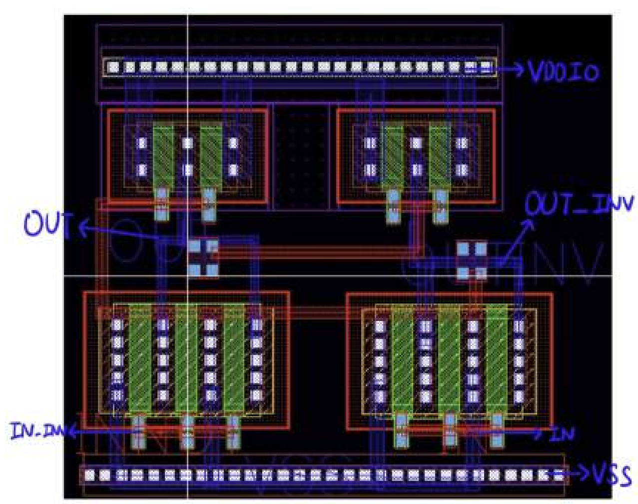

Figure 29: **Layout of Level Shifter**

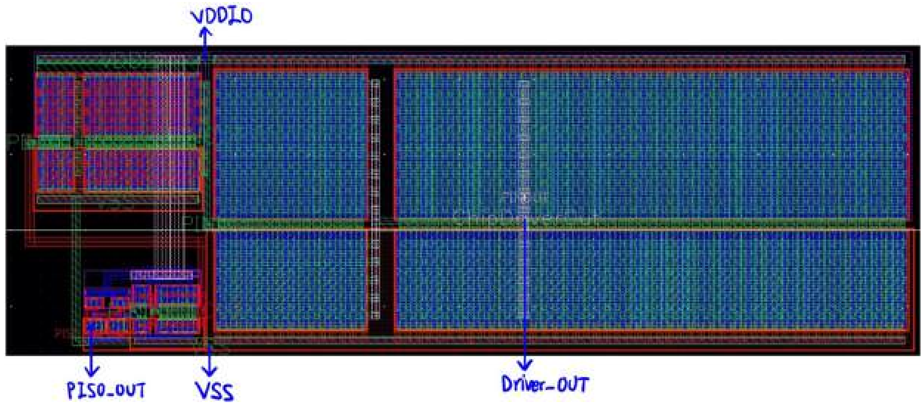

Figure 30: **Layout of Off Chip Driver**

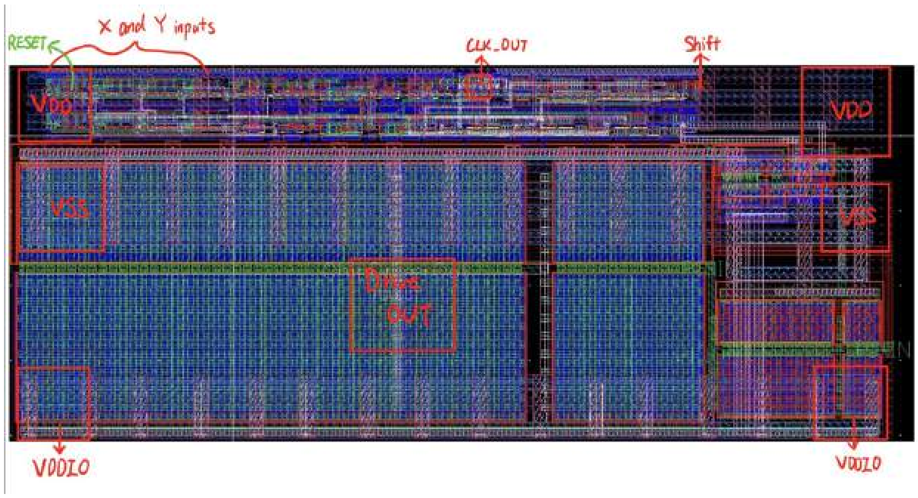

Figure 31: **Layout of Entire Image Comparator.** Red square regions correspond to bondpads.

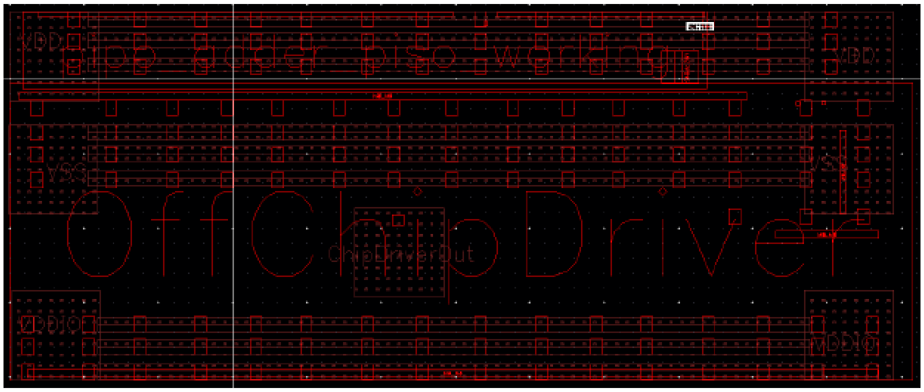

Figure 32: **Layout of Bondpad of Entire Image Comparator for Comparison.**

The layout strategy for this project focuses on achieving functionality, performance, and power efficiency while adhering to design specifications. Below is a detailed approach to the layout process, including the completed modules.

5.1 Hierarchical Design Approach

1. **Top-Level Organization**: The design is structured hierarchically, with the layout divided into functional blocks such as the parallel-in parallel-out (PIPO) and parallel-in serial-out (PISO) shift registers, clock divider, adder modules (half and full adders), level shifter, and the off-chip driver. These modules are developed independently to be integrated into a cohesive top-level design.

2. **Block-Level Verification**: Each block is verified individually to ensure correctness, facilitating targeted debugging and optimization.

3. **Integration**: Once verified, these modules are integrated with attention to logical signal flow and minimal interconnect delays, ensuring seamless interaction between components.

5.2 Power and Ground Routing

1. **High-Level Metal Usage**: High-level metals (Metal 9 for VDD and VDDIO) are used for power distribution, ensuring low resistance and spatially uniform voltage delivery.

2. **Horizontal and Vertical Routing**: VDD and VDDIO are routed horizontally along the top and bottom edges of the layout, while power is distributed across the chip by vertical buses. VSS nets are routed horizontally on Metal 8, branching out from bond pads for uniform ground distribution.

3. **Subcircuit Connections**: Each module is connected to the top-level power and ground mesh through lower-level metals, providing robust power delivery to all components.

5.3 Clock Distribution

1. **Centralized Clock Network**: The clock signal is distributed from a central point to all registers, with Metal 5 chosen for its balanced resistance and capacitance properties.

2. **Clock Buffers**: Buffers are strategically placed to minimize skew and ensure high-speed signal integrity at the target frequency of 2.4 GHz.

3. **Clock Divider Integration**: A clock divider generates the lower-frequency clock (CK/4), ensuring synchronized operation of modules that require different timing signals.

5.4 Component Placement

1. **Critical Modules**: Key modules, including the XOR gates, adders (half adders and full adders integrated into a complete adder unit), and the PISO shift register, are placed close together to minimize delays and interconnect parasitics.

2. **Logical Data Flow**: The placement supports a logical flow of data, beginning with the input from the PIPO shift register, through the adder, and into the PISO shift register for serialization.

3. **Thermal and Noise Management**: Modules generating significant heat or noise, such as the level shifter and off-chip driver, are spaced appropriately to maintain thermal and electrical stability.

5.5 Signal Routing

1. **Layer Management**: We utilized signal routing to avoid interference with power and clock networks through the use of lower-level metals. Horizontal and vertical routing is alternated across layers to maintain orthogonality and reduce routing complexity.

2. **Critical Path Optimization**: High-speed paths, particularly those between the adder and PISO shift register, are kept short and direct to meet timing constraints.

5.6 Area and Power Optimization

1. **Compact Layout**: The layout is compact but provides sufficient spacing for routing and thermal management. This ensures efficient use of chip area while meeting manufacturability standards.

2. **Transistor Sizing**: Transistor sizes are optimized to balance power consumption and performance, ensuring functionality across all process corners (TT, SS, SF, FS, FF).

3. **Power Domain Separation**: The core circuitry (VDD) operates at 1.1 V, while the output driver (VDDIO) is powered at 1.8 V. This separation minimizes power loss and ensures compatibility with the off-chip load.

5.7 Verification and Debugging

1. **Iterative Testing**: Each module is individually tested before integration. C-extracted simulations are used for initial testing, followed by RC-extracted simulations to account for parasitic effects in the final layout.

2. **Process Corner Validation**: The design is verified across all process corners using the provided testbench to ensure robust operation under varying conditions.

3. **Latency Measurement**: Latency is measured in clock cycles from the first rising edge of CK after Reset is de-asserted, ensuring the design meets timing requirements.

5.8 Output Driver Considerations

1. **Level Shifter Design**: A level shifter is used to convert the core voltage levels (1.1 V) to the higher output driver levels (1.8 V).

2. **Driver Sizing**: The off-chip driver is sized to efficiently drive the 10 pF load, achieving a swing between 0.1·VDDIO and 0.9·VDDIO within a single clock period.

5.9 Final Layout Considerations

1. **Top-Level Integration**: The final layout integrates all modules, including the PIPO and PISO registers, clock divider, adder (half, full, and combined), level shifter, and off-chip driver, into a cohesive and efficient design.

2. **Routing Compliance**: The layout adheres to DRC rules for spacing, metal width, and via placement to ensure manufacturability and reliability.

3. **Visualization**: The top-level view highlights the hierarchical structure and efficient organization of the layout.

6 Workload Distribution

Finally, the workload distribution for our group is shown below.

- **Muye**: Clock Divider and Shift Signal Generator, PIPO and PISO Module
- **Kadin**: Level Shifter, Off Chip Driver, PIPO Module, Entire Layout Connections
- **Ananya**: Adder Module, Final Report
- **Ben**: Final Report, Analysis of Data